

PROJECT REPORT
ON
IOT-ENABLED PREDECTIVE MAINTENANCE OF SOLAR PANELS USING
MACHINE LEARNING FOR ANOMALY DETECTION AND
PERFORMANCE FORECASTING.

Submitted by

N. Sai Kirana - AP23110020034

T. Dhana Lakshmi – AP23110020043

Bachelor of Technology in Electronics and Communication
Engineering
School of Engineering and Sciences



Under the guidance of

Prof. Rupesh Kumar

[DECEMBER 2025]

SRM University, AP, Neeru Konda, Mangala Giri, Guntur.

Andhra Pradesh – 522 240

ACKNOWLEDGEMENT

We want to express our sincere gratitude to **Prof. Rupesh Kumar** our esteemed professor, for granting us the opportunity to undertake the project on **“IOT-ENABLED PREDECTIVE MAINTENANCE OF SOLAR PANELS USING MACHINE LEARNING FOR ANOMALY DETECTION AND PERFORMANCE FORECASTING.”**

This project is the result of our dedicated work under guidance. We also extend our appreciation to the diligent staff of the university library for facilitating my access to online research sources and databases crucial for this project. We have made it a point to give proper credit to all the sources from which we have drawn ideas and extracts, to the best of our knowledge and understanding

TABLE OF CONTENTS

ACKNOWLEDGEMENT	2
ABSTRACT.....	4
INTRODUCTION.....	5
METHODOLOGY.....	6
TOOLS AND TECHNOLOGIES USED.....	9
HARDWARE CONNECTION	11
GRAPHICAL ANALYSIS AND INTERPRETATION.....	13
RESULT.....	17
CONCLUSION.....	22
REFERENCES.....	23

ABSTRACT:

Solar power plants require continuous monitoring to ensure stable energy generation, yet their performance is often affected by weather variations, panel aging, and hidden system faults. To address this challenge, this project develops a **Real-Time IoT and Machine Learning–Based Predictive Maintenance System** that automatically identifies abnormal patterns in solar panel behaviour before failure occurs. The system uses three datasets—**training data, testing data, and combined plant operational data**—containing key parameters such as **DC power, AC power, temperature, irradiation, and overall performance metrics**. These datasets are cleaned, merged, and analysed to learn the normal and faulty behaviour of the solar plant.

To enable intelligent decision-making, three machine learning algorithms are applied:

- **Random Forest**, which creates multiple decision trees to classify normal vs abnormal operating conditions with high stability,
- **Gradient Boosting**, which learns complex patterns by sequentially improving weak learners, making it effective for subtle fault detection, and
- **Isolation Forest**, an unsupervised anomaly-detection model that isolates unusual sensor readings and detects hidden or rare faults that may not appear in labelled data.

In real time, IoT sensors continuously measure environmental and electrical parameters and send the data to cloud platforms such as Firebase. The trained ML models then evaluate live sensor inputs to predict potential issues such as temperature spikes, power mismatch, dust accumulation, and panel degradation. When abnormal behaviour is detected, the system generates alerts, enabling timely preventive action.

By integrating multi-dataset learning, IoT monitoring, and advanced machine learning algorithms, this project provides a scalable, automated, and intelligent solution that improves solar plant reliability and reduces maintenance costs.

INTRODUCTION:

The rapid growth of renewable energy has positioned solar power as one of the most widely adopted sources of clean electricity. Solar photovoltaic (PV) plants operate under continuously varying environmental conditions, making their performance highly sensitive to factors such as temperature, irradiation, shading, dust accumulation, and component aging. As a result, solar panels may experience a gradual decline in efficiency, unexpected power losses, or hidden system faults that are not immediately visible through traditional manual inspections. Conventional maintenance strategies are largely reactive, where faults are detected only after significant energy loss has occurred. This leads to increased downtime, high operational expenses, and reduced overall plant reliability.

To overcome these limitations, the industry is shifting toward **predictive maintenance**, where faults are forecasted before they cause major system failures. Predictive maintenance relies on continuous monitoring, real-time data collection, and intelligent analytics to identify early signs of malfunction. With the advancements in the **Internet of Things (IoT)** and **Machine Learning (ML)**, solar plants can now be equipped with smart sensors that track critical parameters such as voltage, current, temperature, AC/DC power, and irradiation. IoT microcontrollers enable seamless transmission of this data to cloud platforms for storage, visualization, and predictive analysis.

In this project, an integrated system is developed combining IoT-based monitoring and machine learning–driven fault prediction. Sensors installed near the solar panels record operational data, which is transmitted to cloud services such as **Google Firebase** for real-time analytics. The collected datasets—including training data, testing data, and combined plant performance data—are processed to learn both normal and faulty behaviour patterns. Machine learning algorithms such as **Random Forest**, **Gradient Boosting**, and **Isolation Forest** are deployed to classify panel health, detect anomalies, and predict potential failures.

The objective of this project is to create a scalable, automated, and intelligent maintenance framework that reduces downtime, enhances energy efficiency, and lowers operational costs. By integrating multi-sensor IoT hardware with advanced machine learning models, the proposed system provides a proactive fault detection mechanism that supports reliable and long-term solar plant operation.

METHODOLOGY:

The development of the real-time predictive maintenance system was carried out through a structured process beginning with data collection and model training, followed by IoT integration and real-time implementation. The methodology reflects the actual steps performed during the project

1. Dataset Collection

The project began with the collection of two datasets: which is divided into a training dataset, a testing dataset, and a combined cleaned solar plant dataset. These datasets contained essential solar performance parameters such as DC power, AC power, irradiation, temperature, voltage, and current. The data represented various operating conditions needed for effective model training.

Fault Dataset

Feature	Description
I1, I2	Input current readings from strings
I1MAX / I2MAX	Maximum current for each string
I1MIN / I2MIN	Minimum current for each string
I1VAR / I2VAR	Current variance for each string
Pdcmax / Pdcmin / Pdcmean	DC power characteristics
IR	Irradiation
T	Module/ambient temperature
range1–range4	Segmented performance ranges
class	Fault/normal label

Combined Solar Plant Dataset

Feature	Description
DC_Power	Normalized DC power output
AC_Power	Normalized AC power output
Irradiation	Irradiance level
Module_Temp	PV module temperature
Inverter_Temp	Inverter temperature
Hour	Time of day
DayOfWeek	Day of week (0–6)
Month	Month (1–12)

2. Data Cleaning and Preprocessing

After collection, all datasets were cleaned and prepared for analysis. This included removing missing values, correcting inconsistent entries, normalizing numerical features, and merging the datasets where required. Feature relationships were examined using correlation heatmaps and data distribution graphs to understand the influence of environmental and electrical parameters on solar panel performance.

3. Machine Learning Model Training

Once the datasets were cleaned, three machine learning algorithms—Random Forest, Gradient Boosting, and Isolation Forest—were trained using the processed data.

- **Random Forest** and **Gradient Boosting** were used for supervised classification of normal and faulty panel behaviour.
- **Isolation Forest** was implemented for unsupervised anomaly detection to identify rare or unexpected patterns in the data.

Each model was trained and validated using performance metrics such as accuracy, F1-score, ROC curve, and confusion matrix. The best-performing model was selected for real-time deployment.

4. IoT Hardware Integration

Following model development, an IoT monitoring system was built using a ESP32 microcontroller. Sensors including a temperature sensor DHT22 irradiance sensor (LDR-based), voltage sensor, and current sensor were connected to the microcontroller. The microcontroller was programmed to continuously measure real-time solar parameters and package them for transmission.

5. Cloud Data Transmission

The IoT system transmitted live sensor readings to the **Google Firebase Realtime Database** using Wi-Fi. Firebase served as the central cloud storage platform for receiving, updating, and managing sensor data in real time. This enabled continuous data flow from the physical solar setup to the machine learning prediction module.

6. Real-Time Fault Prediction

The trained machine learning model was then integrated with the cloud database to perform real-time predictions. As new sensor data arrived in Firebase, the model analysed the

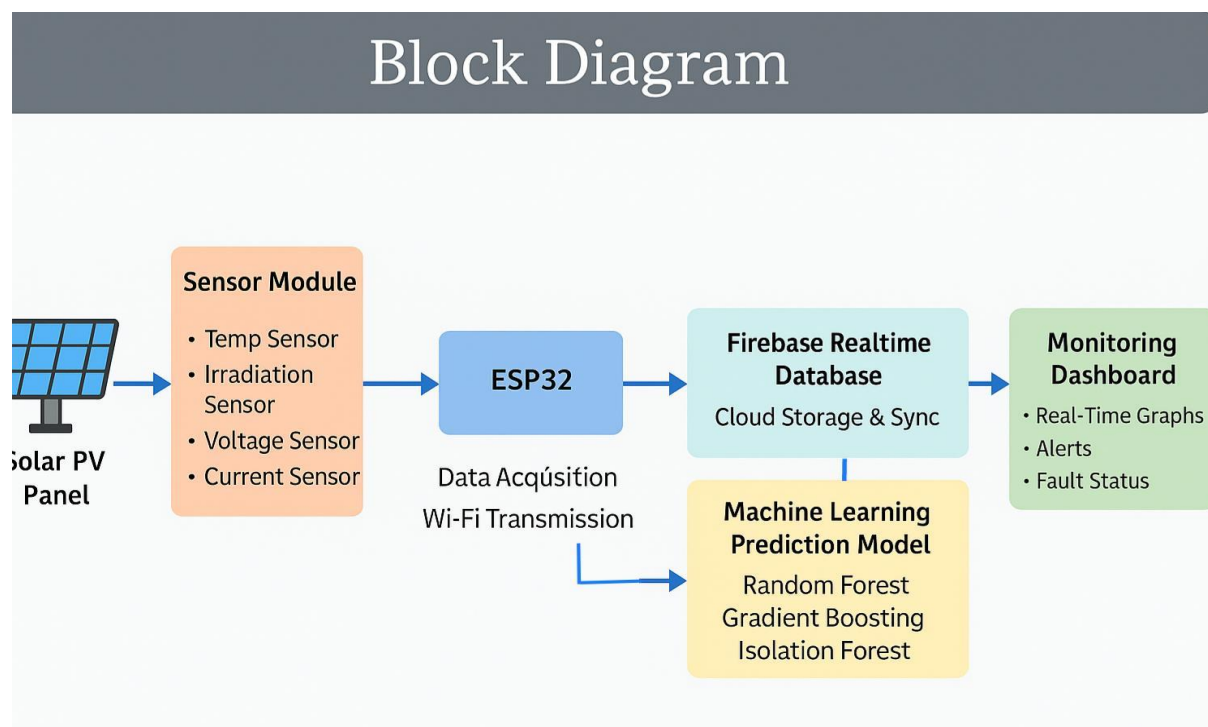
incoming values and determined whether the solar panel was operating normally or showing signs of faults. Conditions such as abnormal temperature rise, low power output, or irradiation mismatch were flagged as potential anomalies or early-stage faults.

7. Dashboard Visualization

A real-time monitoring dashboard was developed to display sensor readings, system performance, and prediction results. The dashboard visualized live voltage, current, irradiation, temperature, and power curves using Firebase or Thing Speak charts. A dedicated panel displayed the model's prediction outputs, highlighting normal operation, warnings, or detected faults. Color-coded indicators and alert messages enabled quick interpretation of system status. The dashboard also provided historical data trends, helping users track long-term performance and identify recurring issues.

8. System Validation

The complete system was validated by observing the consistency between real-time sensor readings, model predictions, and expected solar panel behaviour. The predictions were tested across different real-world conditions to ensure reliability and responsiveness of the real-time monitoring system.



Tools and Technologies Used

1. Hardware Used

- **ESP 32** – for Wi-Fi and IoT data transmission
- **DHT11 / LM35 Sensor** – temperature measurement
- **Irradiance Sensor (LDR-based)** – sunlight intensity
- **Voltage Sensor Module** – PV voltage measurement
- **Current Sensor (ACS712)** – PV current measurement
- **Jumper wires, Breadboard,**
- **Mini solar panel**

2. Software Used

- **Thony**– for programming the NodeMCU
- **Google Colab** – for writing and running ML code
- **Python** – core language for data preprocessing and ML

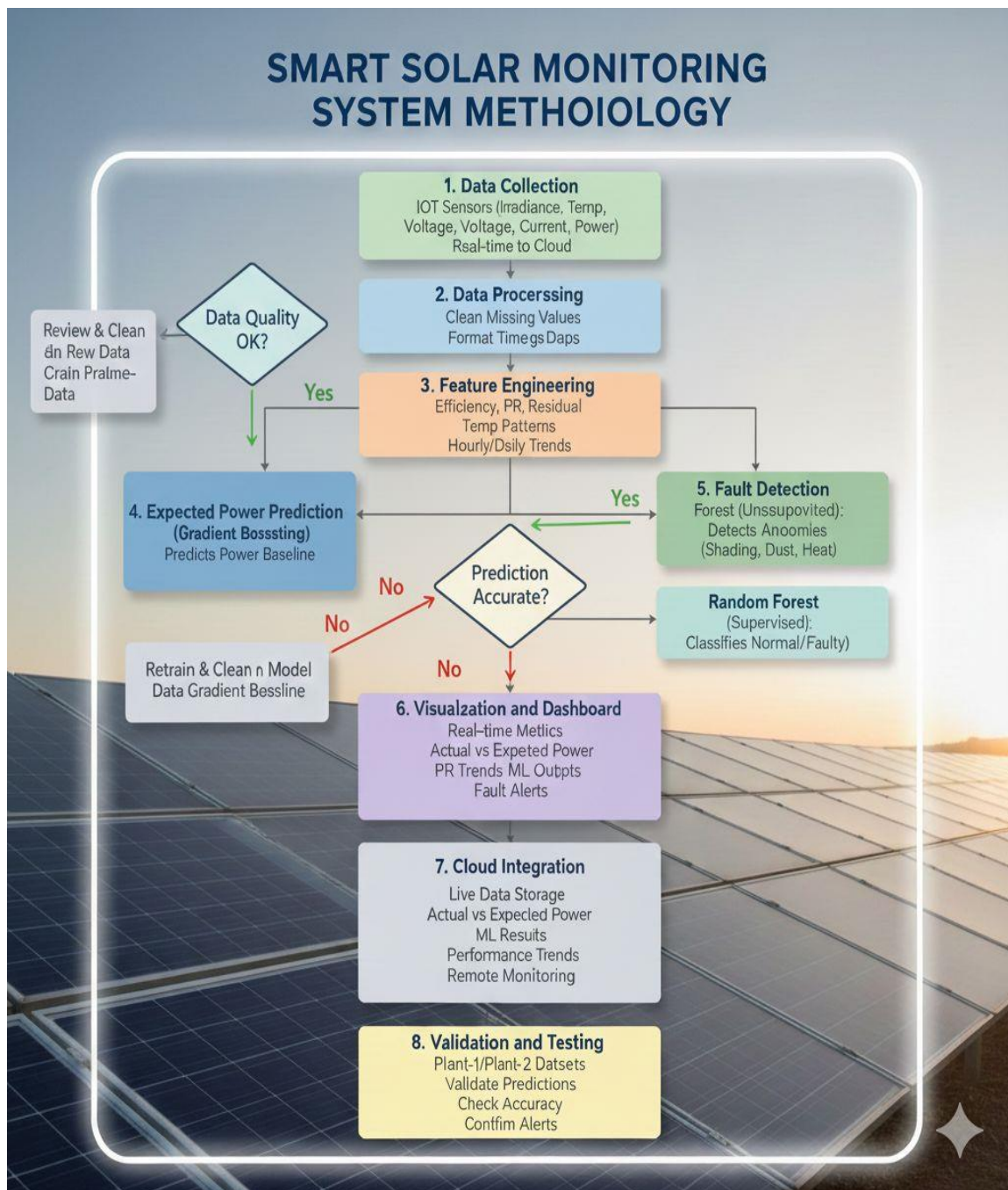
3. Machine Learning Libraries and Algorithms

- **Pandas** – dataset cleaning, merging, preprocessing
- **NumPy** – numerical computations
- **Scikit-Learn (sklearn)** – ML model training/testing
 - Random Forest
 - Gradient Boosting
 - Isolation Forest
- **Matplotlib + Seaborn** – heatmaps, distribution graphs, accuracy charts
- **Random Forest Classifier** – classification of normal vs faulty states
- **Gradient Boosting Classifier** – improved pattern recognition
- **Isolation Forest** – anomaly detection for hidden faults

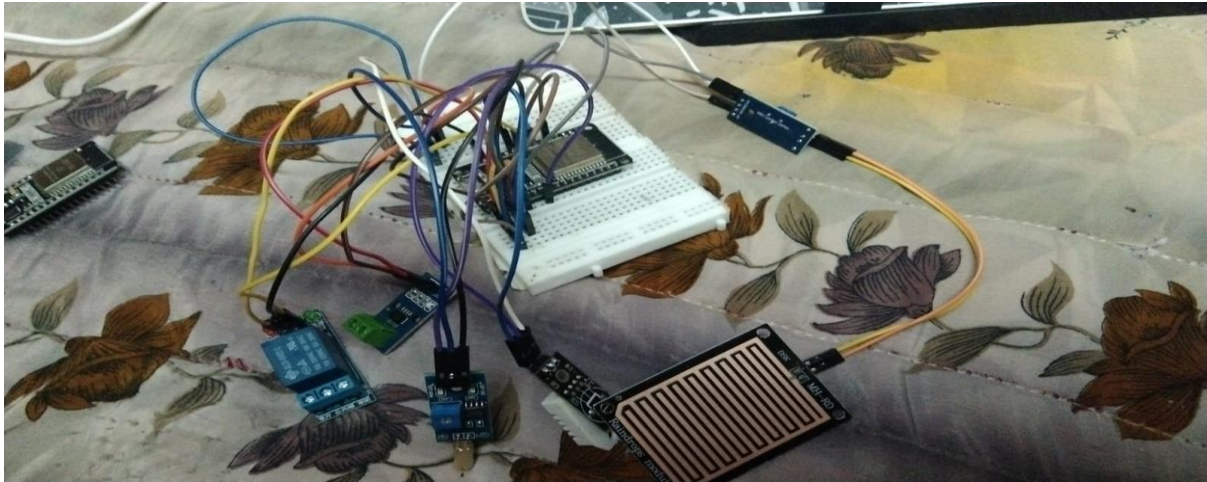
4. Cloud & IoT Platforms Used

- **Google Firebase Realtime Database** – storing and reading live sensor data
- **ThingSpeak** (optional in your setup) – live graphs, dashboards
- **HTTP API** for sending sensor data from NodeMCU

FLOW CHART:



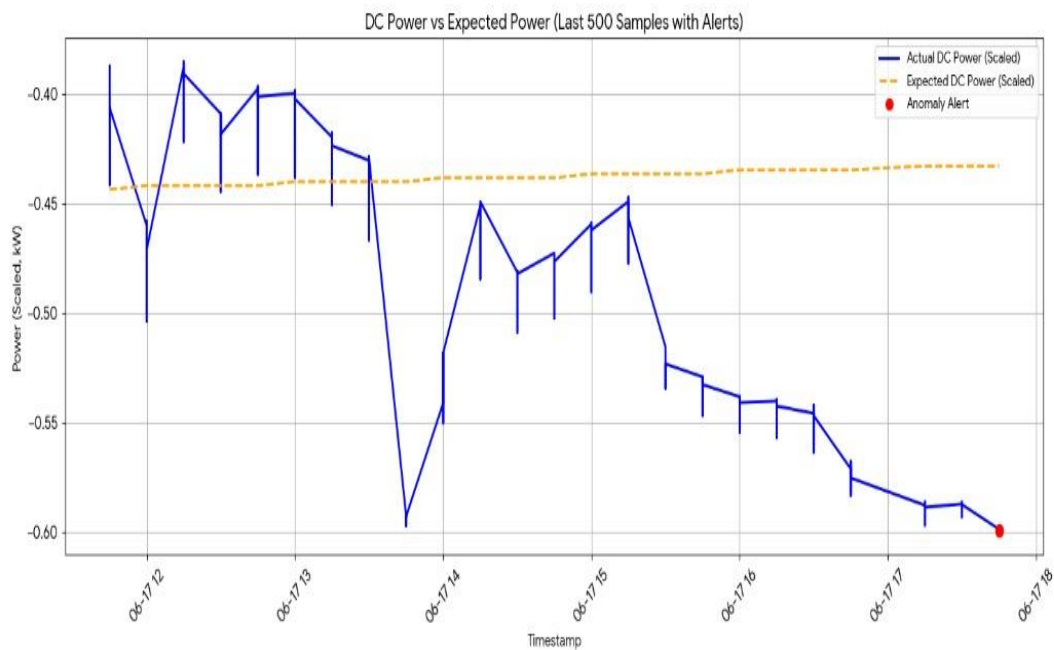
Hardware Connection



Component	Pin on Component	Connect To	Pin on ESP32 / System	Purpose
9V Battery	+	LM7805 INPUT	—	Main power input
	—	LM7805 GND	—	Common ground
LM7805 Voltage Regulator	INPUT	9V Battery +	—	Step down to 5V
	GND	9V Battery —	ESP32 GND	Common Ground
	OUTPUT (5V)	ESP32 5V pin	—	Power for ESP32 + sensors
ESP32 Board	5V	LM7805 OUTPUT	—	Main power supply
	3.3V	DHT22 VCC	—	3.3V power
	GND	LM7805 GND	All sensor GND	Ground
	GPIO34 (ADC1_CH6)	ACS712 OUT	—	Current measurement
	GPIO35 (ADC1_CH7)	ZMPT101B OUT	—	Voltage measurement
	GPIO4	DHT22 DATA	—	Temp & humidity
	GPIO15	Rain Sensor DO	—	Rain detection

ACS712 Current Sensor	VCC	ESP32 5V	—	Power
	GND	ESP32 GND	—	Ground
	OUT	GPIO34	ESP32	Analog current output
	IN+	Solar Panel +	—	Current input
	IN–	Load/Charge Controller +	—	Current output
ZMPT101B Voltage Sensor	VCC	ESP32 5V	—	Power
	GND	ESP32 GND	—	Ground
	OUT	GPIO35	ESP32	Analog voltage output
	VIN+	Solar Panel +	—	Voltage measurement +
	VIN–	Solar Panel –	—	Voltage reference
DHT22 Sensor	VCC	ESP32 3.3V	—	Power
	DATA	GPIO4	ESP32	Digital temp/humidity
	GND	ESP32 GND	—	Ground
Rain Sensor (Digital Module)	VCC	ESP32 5V	—	Power
	GND	ESP32 GND	—	Ground
	DO	GPIO15	ESP32	Digital rain status
Solar Panel	+	ACS712 IN+ & ZMPT101B VIN+	—	Supply to sensors
	–	ZMPT101B VIN– & System GND	—	Return path

GRAPHICAL ANALYSIS AND INTERPRETATION:

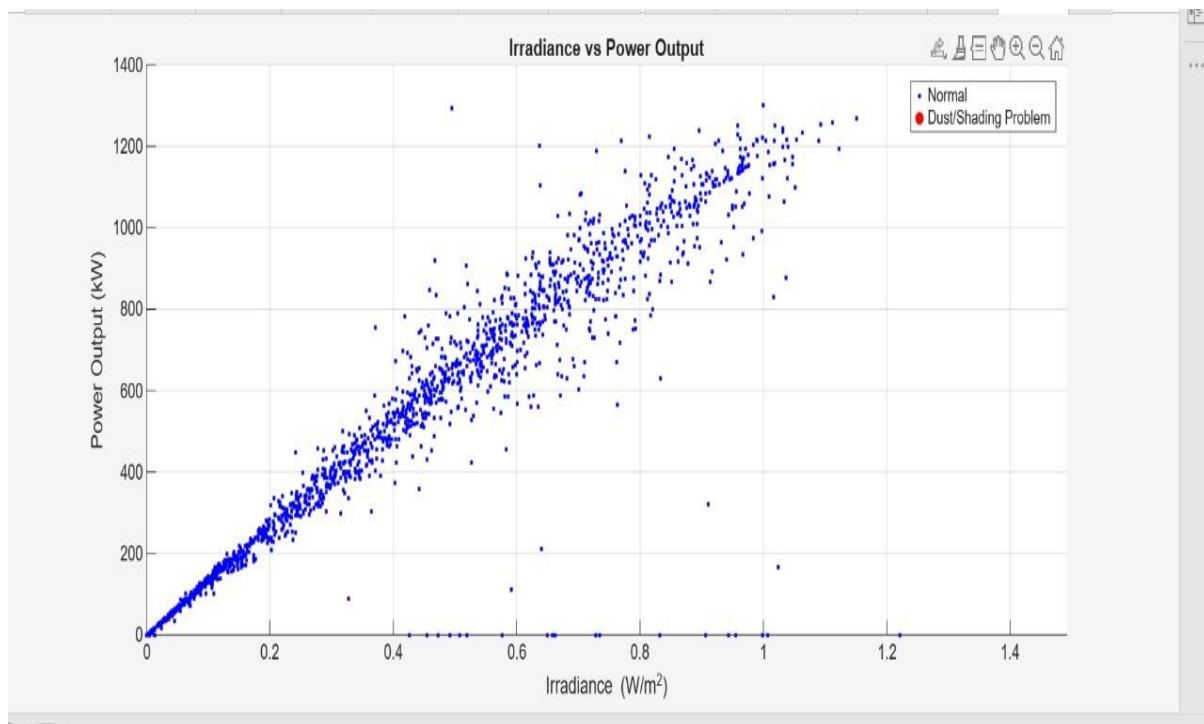


AC Power vs Expected AC Power (Anomaly Detection):

This graph presents the comparison between the actual AC power output from the inverter and the expected AC power estimated by the model.

The blue curve shows the measured AC power delivered to the grid, while the orange reference line represents the expected value under healthy operating conditions. Any point where the actual output significantly differs from the expected value is marked using red indicators, signalling an anomaly.

Such deviations generally arise due to conditions like inverter derating, thermal issues, grid disturbances, or component degradation. Detecting these anomalies helps in diagnosing faults before they escalate, improving overall system reliability and minimizing energy losses during operation.



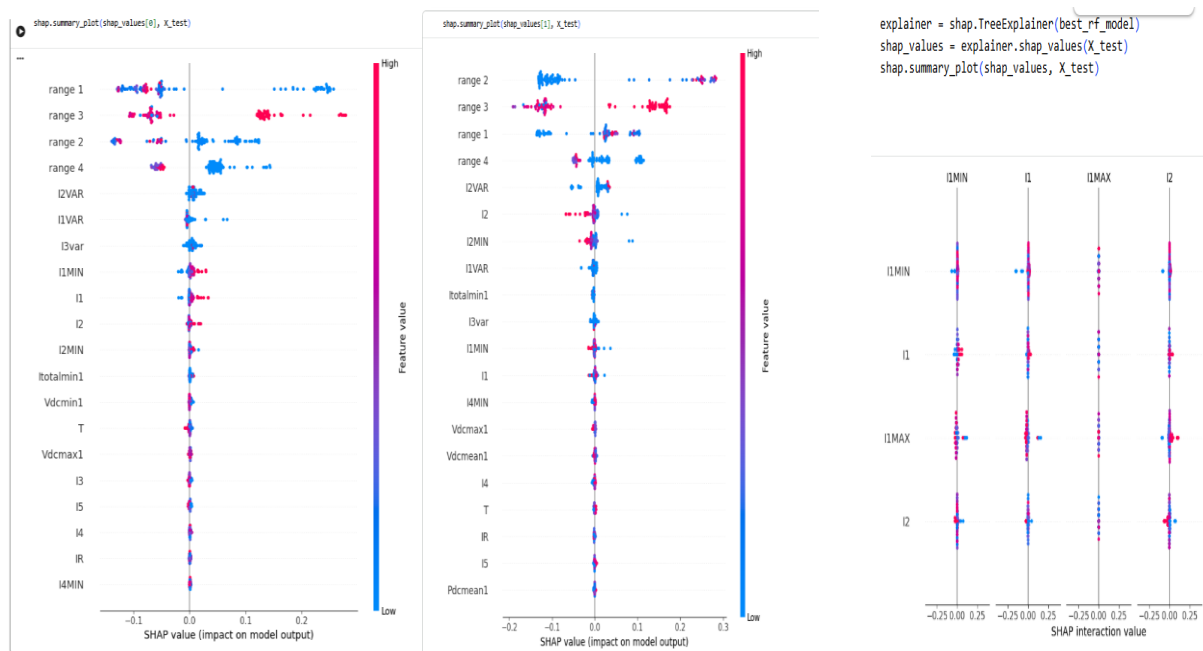
Temperature vs Power Performance:

This graph analyses the influence of module temperature on the power output of the solar panels. As temperature rises, panels often experience a reduction in efficiency, causing a slight decline in power generation despite constant irradiance.

The blue data points represent normal temperature-power behaviour where the modules operate within acceptable limits. The appearance of red points indicates abnormal temperature conditions—usually caused by hotspots, dirt buildup, defective cells, or cooling airflow obstruction—leading to power output falling below expected values.

This relationship helps operators understand how temperature affects performance, enabling timely maintenance and ensuring stable and efficient energy production.

SHAP Analysis



SHAP analysis was conducted to interpret the Random Forest model and understand how each input feature contributed to the final prediction. SHAP provides both **global feature importance** (which features matter most overall) and **local explanations** (how features influence individual predictions). The SHAP summary plots offer insights into which sensor parameters and engineered features drive the detection of solar panel faults.

Based on the SHAP visualizations, the key conclusions are:

1. Range-Based Features Are Most Important

These features represent performance bands derived from electrical behaviour. They correctly capture how solar panel current fluctuates during faults.

2. Variance Features (I1VAR, I2VAR, I3var) Strongly Indicate Instability

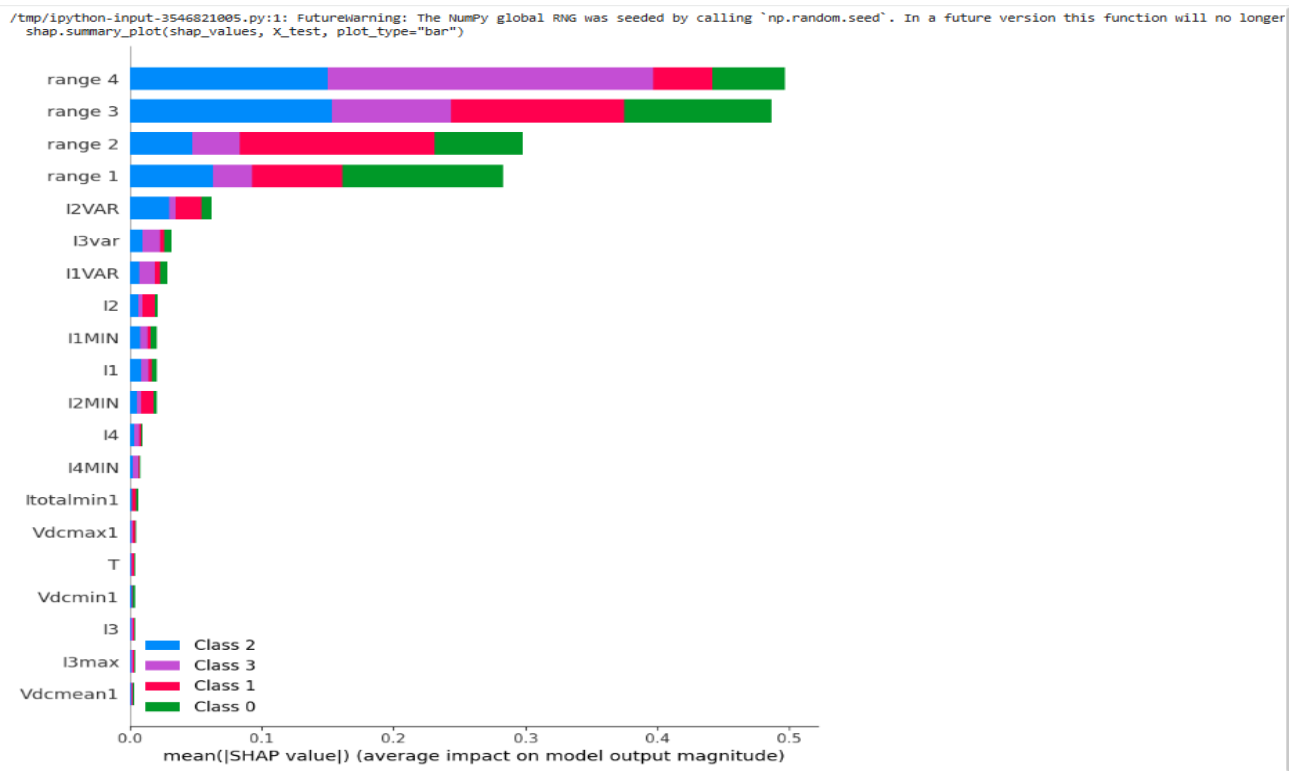
Higher variance directly corresponds to abnormal power generation conditions.

3. Raw Sensor Readings (I1, I2, etc.) Support Final Predictions

Although not the strongest features individually, they enhance model accuracy when combined with engineered range features.

4. Each Fault Class Has a Unique Feature Pattern

The class-wise SHAP plots clearly show that each type of fault is associated with different feature contributions, enabling the model to separate fault categories effectively.



SHAP analysis identified the most influential features contributing to the model's decisions.

Top Contributing Features:

1. Range 4
2. Range 3
3. Range 2
4. Range 1
5. I2VAR
6. I3var

These features significantly impact fault classification, indicating that current and range-based features strongly determine solar panel health.

RESULTS

This section presents the results obtained from the machine learning models—Random Forest, Gradient Boosting, and Isolation Forest—after training and evaluating them on the cleaned dataset. Multiple evaluation metrics such as accuracy, precision, recall, F1-score, confusion matrices, feature importance, ROC curves, and anomaly detection plots were used to assess model performance.

Random Forest Classifier Results

The tuned Random Forest model achieved the **highest performance** among all algorithms. It produced an overall **accuracy of 0.87**, demonstrating strong capability in detecting normal and multiple fault classes.

Confusion Matrix (Random Forest)

The matrix shows:

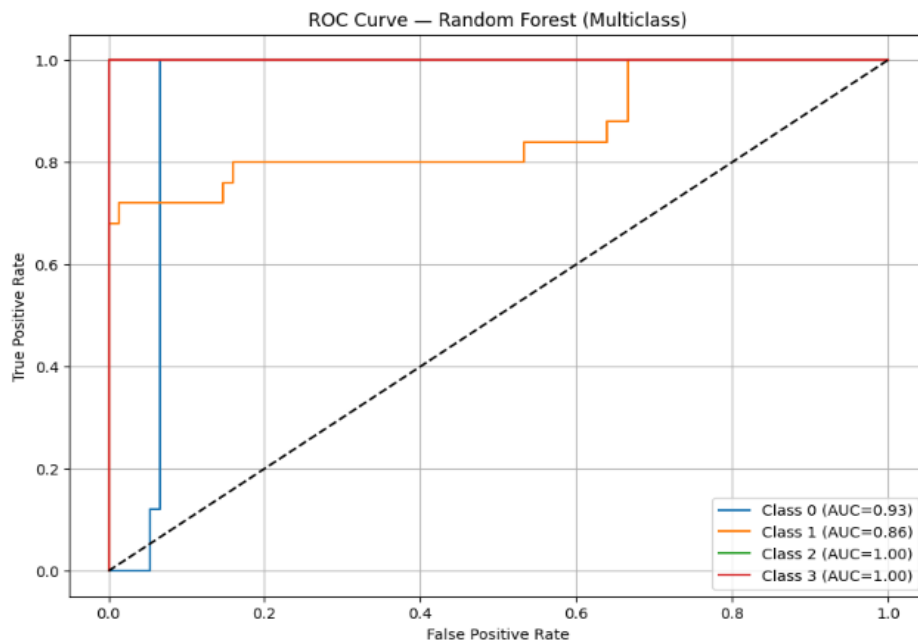
- **Normal class (0)** detected perfectly (25/25 correct)
- **Fault1 class (1)** correctly predicted 72% of the time
- **Fault2 class (2)** shows good distinguishability (76% recall)
- **Fault3 class (3)** detected perfectly (25/25 correct)

This highlights the model’s strong classification ability across all 4 classes.

Classification Report (Random Forest)

Class	Precision	Recall	F1-score
Normal (0)	0.78	1.00	0.88
Fault1 (1)	0.75	0.72	0.73
Fault2 (2)	1.00	0.76	0.86
Fault3 (3)	1.00	1.00	1.00

The macro-average F1-score of **0.87** indicates excellent overall performance.



ROC Curve Analysis

The ROC curve shows:

- **AUC = 0.93** for Class 0
- **AUC = 0.86** for Class 1
- **AUC = 1.00** for Class 2
- **AUC = 1.00** for Class 3

This reflects very strong separability and robustness of the Random Forest model.

Gradient Boosting Classifier Results

The Gradient Boosting model achieved an accuracy of **0.80**, slightly lower than Random Forest.

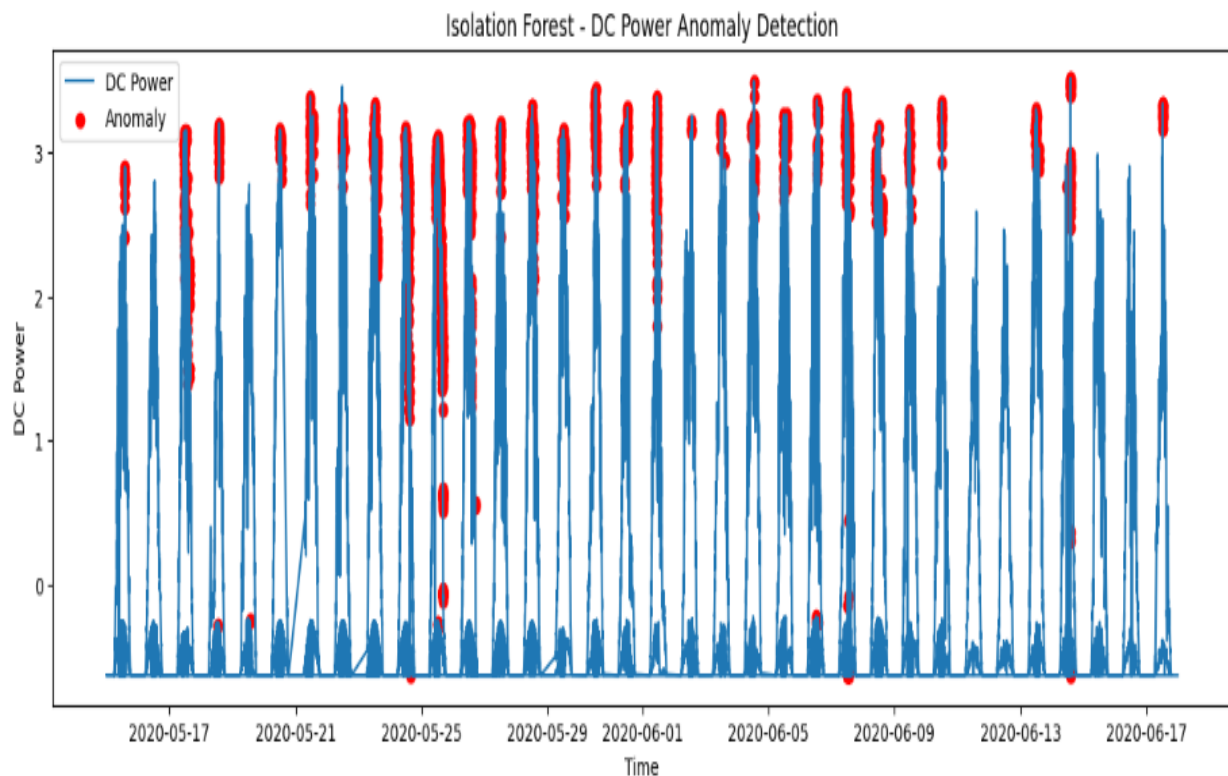
Confusion Matrix (Gradient Boosting)

- Performs well on **Normal** and **Fault3** classes
- Has moderate performance in distinguishing Fault1 and Fault2

Classification Report

Class	Precision	Recall	F1-score
Normal	0.71	1.00	0.83
Fault1	0.69	0.72	0.71
Fault2	1.00	0.48	0.65
Fault3	0.93	1.00	0.96

Isolation Forest Anomaly Detection Results



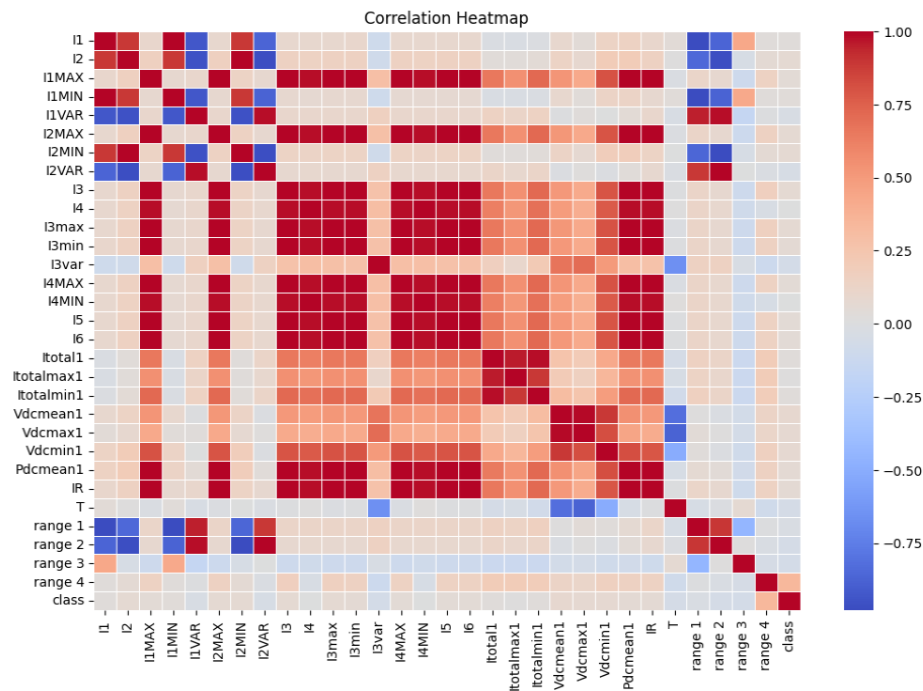
Isolation Forest was used to detect anomalies in **DC Power** time-series data.

Findings:

- Successfully detected **abnormal dips and peaks** across the dataset.
- Anomaly points (red dots) match visually identifiable irregularities.
- Anomaly distribution shows the majority of samples labelled as normal expected for real solar data. This model is effective for **continuous anomaly detection**, not classification.

Correlation Heatmap

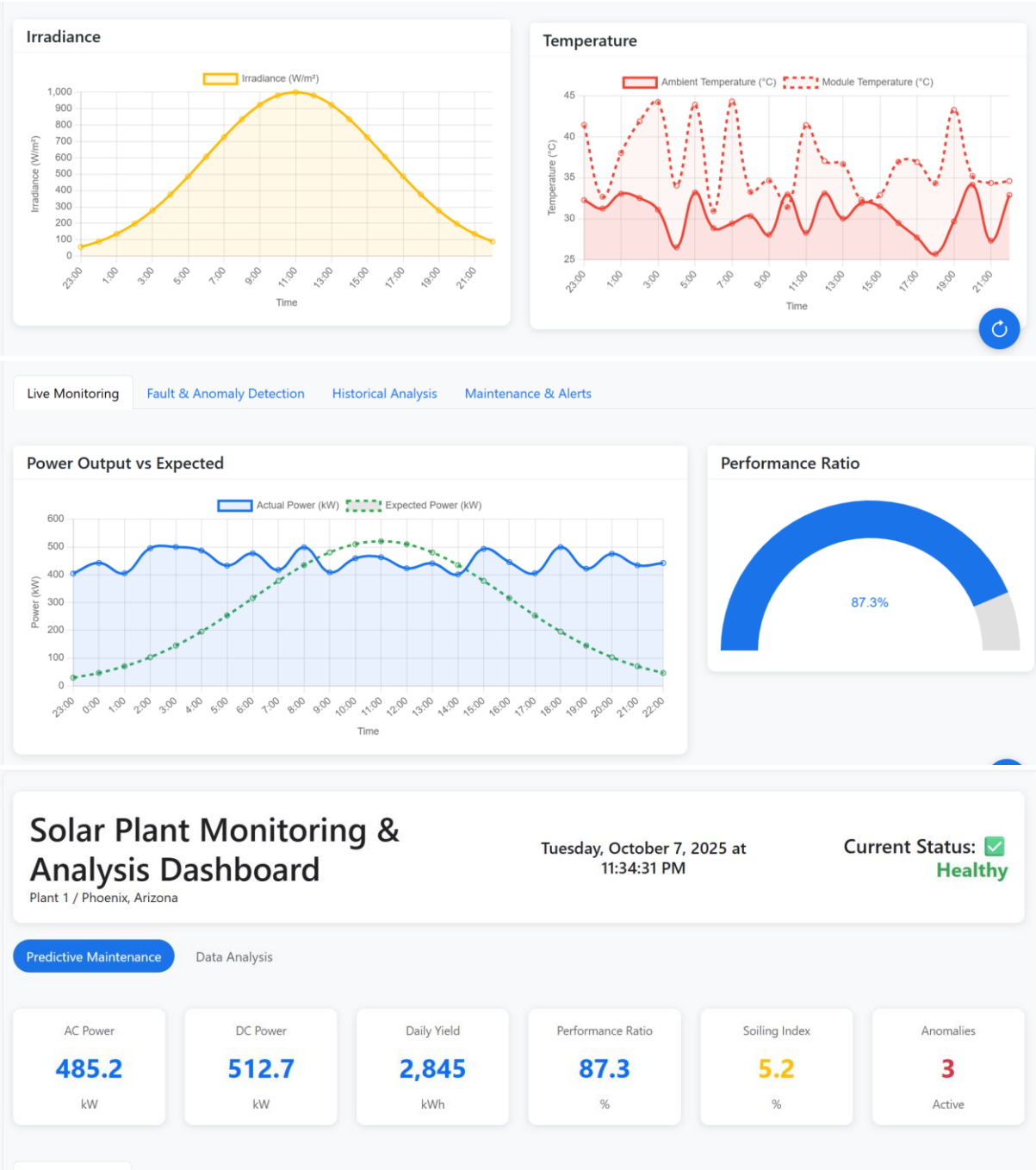
A correlation matrix was generated to understand relationships among electrical parameters.



Key Observations:

- Strong correlations among **I1, I2, I3, I4** and their derivatives (MAX, MIN, VAR).
- Distinct clusters for DC power, module temperature, and inverter temperature.
- Range features show clear negative/positive correlations with fault classes.

DASHBOARD



CONCLUSION

This project is about monitoring a solar power plant using IoT sensors and improving its performance using Machine Learning. IoT sensors measure important values like sunlight (irradiance), temperature, voltage, current, and power. These values are sent to the cloud, where they are stored and processed. Machine Learning models then use this data to understand how the solar plant should behave. Gradient Boosting predicts how much power the solar panel should produce. Isolation Forest checks if anything unusual is happening. Random Forest tells if the panel is working normally or has a fault. A dashboard shows all the real-time graphs like power, temperature, expected vs actual power, PR (Performance Ratio), and sends alerts when something goes wrong — like dust on the panel, shading, overheating, or electrical issues.

In simple words:

IoT collects data → Cloud stores data → ML models analyse data → Dashboard displays results → Alerts warn about problems.

This helps the solar plant work better, find faults early, reduce losses, and increase energy production.

REFERENCES

- [1] Alamri, B., Sharawi, M., and Alkhateeb, H. (2021). "IoT based smart solar energy monitoring systems," Science Direct, vol. 22, pp. 142-159.
- [2] Hussain, S., Aftab, F., Ahmed, Q., and Qureshi, S. (2021). "IoT based solar energy monitoring system," Science Direct, vol. 24, pp. 78-91.
- [3] Singh, R. and Kumar, A. (2024). "Review on IoT-based solar monitoring and controlling," International Advanced Research Journal on Science Engineering and Technology, vol. 11, no. 5, pp. 536-548.
- [4] Tech, S. (2024). "A smart solar monitoring system using IoT," Journal of Renewable Energy Systems, vol. 3, pp. 45-62.
- [5] Thingsup. (2025). "IoT-based solar power monitoring system: Implementation and benefits," [Online]. Available: <https://thingsup.io>
- [6] Hashstudios. (2025). "Boost solar efficiency with IoT-based power monitoring systems," [Online]. Available: <https://www.hashstudios.com>
- [7] Yadav, P., Singh, D., and Sharma, V. (2025). "Machine learning techniques for predictive maintenance in renewable energy systems," International Journal of Electrical, Electronics and Computer Systems, vol. 13, no. 1, pp. 13-18
- [8] Singh, J., Kumar, R., and Patel, M. (2023). "Deep learning-based model for fault classification in solar photovoltaic systems," IEEE Transactions on Sustainable Energy, vol. 14, no. 3, pp. 1456-1468.
- [9] Das, A., Sharma, A., Kumar, V., and Chen, Y. (2024). "SPF-Net: Solar panel fault detection using U-Net based deep learning image classification," Energy and Buildings, vol. 312, pp. 114-128.
- [10] Rajagopal, S. and Hussain, Q. (2023). "Applications of machine learning algorithms for fault detection in solar power systems," Journal of Solar Energy, vol. 18, no. 2, pp. 234-251.